

Control work 7 · Data and counting

The Chapter IV assessment, and the four errors it produces every year.

LESSON 89–90

2 × 40 MIN

ALGEBRA 8, CHAPTER IV

STAGE 9 · 12.1–12.3, 15.1

LESSON CLOCK

2'

36'

2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess averages, grouped data, charts, listing and the counting principle.
- Diagnose each lost mark by error type.
- Re-solve every task that was lost.

TEXTBOOK

National	Revision of Chapter IV
Cambridge	Learner's Book pp. 255–270, 318–324
Workbook	Workbook 12.1–12.3

01

Explanation

Lesson 89 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each:

- task 1 · mean, mode, median and range of a small list
- task 2 · read a value from a frequency table
- task 3 · estimate the mean of grouped data
- task 4 · choose and justify a suitable chart
- task 5 · list all outcomes systematically
- task 6 · the multiplication rule
- task 7 · an “at least one” count

Marking note: task 1 loses a mark if the list is not sorted before the median is read.

Lesson 90 — work on mistakes (40 minutes)

THE FOUR ERRORS THIS PAPER PRODUCES

1. Median read from an unsorted list.

Sorting is part of the method, not a nicety.

2. Grouped mean without midpoints — adding the class boundaries, or the frequencies alone.

3. **Adding when the question says “and then”** — the counting rules swapped over.
4. **“At least one” counted directly**, producing double-counted cases instead of *total* — *none*.

Each pupil writes the error number beside every lost mark, then re-solves that task in full.

02 Worked examples

EXAMPLE 1 Find the error: the median of 7, 2, 9, 4, 5 is 9

- ① The list was never sorted.
The middle **of the sorted list** is what the median means.

- ② 2, 4, 5, 7, 9
Sort first.

- ③ $median = 5$
The third of five values.

ANSWER

5

EXAMPLE 2 Find the error: classes 0–10, 10–20 with frequencies 4, 6 give a mean of $(10 + 20) \div 2 = 15$

① Class **midpoints** are needed, weighted by frequency.

② *midpoints 5 and 15*

③
$$\frac{5 \cdot 4 + 15 \cdot 6}{10} = \frac{110}{10} = 11$$

ANSWER

11

03 Interactive model

Show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: the median of 12, 5, 8, 3, 20, 7 is 8 and 3 $\rightarrow 5.5$

① 3, 5, 7, 8, 12, 20

Sort
the
list
first
—
always.

② Six values, so the median is the mean of the 3rd and 4th.

③ $(7 + 8) \div 2$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Mean	O‘rta arifmetik	Среднее арифметическое
Median	Mediana	Медиана
Modal class	Moda sinfi	Модальный класс
Estimated mean	Taxminiy o‘rta	Оценка среднего
Counting principle	Sanashning asosiy qoidasi	Основной принцип счёта
Complementary counting	To‘ldiruvchi hisoblash	Подсчёт через дополнение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The median of 7, 2, 9, 4, 5 is:

For grouped data the estimated mean uses:

3 starters and 4 mains give how many two-course meals?

The best way to count “at least one 6 in two dice throws” is:

add the cases

$36 - 25$

$6 + 6$

$2 \cdot 6$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** Find the mean, mode, median and range of 4, 7, 4, 9, 6

ANSWER mean 6, mode 4, median 6, range 5

2. **Task 2.** Marks 1–5 have frequencies 2, 5, 8, 3, 2. How many pupils?

ANSWER 20

3. **Task 3.** Estimate the mean for 0–10, 10–20, 20–30 with frequencies 3, 5, 2.

ANSWER 14

4. **Task 4.** Which chart shows parts of one whole?

ANSWER A pie chart.

5. **Task 5.** List all two-digit numbers from 1, 2, 3 with repeats.

ANSWER 11, 12, 13, 21, 22, 23, 31, 32, 33

6. **Task 6.** 5 hats and 3 scarves — how many sets?

ANSWER 15

7. **Task 7.** How many of the 8 outcomes of three coin tosses have at least one head?

ANSWER 7

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** Find the mean, mode, median and range of 12, 5, 12, 8, 3

ANSWER mean 8, mode 12, median 8, range 9

2. **Task 2.** Frequencies 4, 6, 7, 3 for scores 1–4. Find the modal score.

ANSWER 3

3. **Task 3.** Estimate the mean for 0–20, 20–40, 40–60 with frequencies 6, 10, 4.

ANSWER 26

4. **Task 4.** Why can a graph starting at 90 rather than 0 mislead?

ANSWER Small differences are stretched so they look large.

5. **Task 5.** List the ways to choose 2 pupils from A, B, C, D.

ANSWER AB, AC, AD, BC, BD, CD

6. **Task 6.** A code has 2 letters (26) then 2 digits. How many codes?

ANSWER $26^2 \cdot 100 = 67\,600$

7. **Task 7.** How many three-digit codes from 1–5 contain at least one 2?

ANSWER $125 - 64 = 61$

Hard · stretch and reason

7 PROBLEMS

1. *The mean of 8 numbers is 12. Two numbers, 5 and 19, are removed. Find the new mean.*

ANSWER $\frac{96 - 24}{6} = 12$

2. *A set of five whole numbers has mean 7, median 6 and mode 5. Find one such set.*

ANSWER 5, 5, 6, 9, 10

3. *Estimate the mean for 0–5, 5–10, 10–15, 15–20 with frequencies 2, 7, 8, 3.*

ANSWER $\frac{5 + 52.5 + 100 + 52.5}{20} = 10.5$

4. *How many four-digit numbers have all digits different and start with an odd digit?*

ANSWER $5 \cdot 9 \cdot 8 \cdot 7 = 2520$

5. *From 8 people, how many ways to pick a chair and a secretary?*

ANSWER 56

6. *How many three-digit codes from 1–6 have at least two equal digits?*

ANSWER $216 - 120 = 96$

7. *Explain why the mean of a grouped table is only an estimate.*

ANSWER Every value in a class is replaced by the midpoint, so the individual values are lost and the total is only approximated.

07 Homework

Homework — 5 problems

Re-solve every task you lost marks on, with the error number written beside it.

1. Re-solve, in full, each task you lost a mark on.
2. Beside each, write the error number (1–4) that caused it.
3. Find the mean, mode, median and range of 6, 9, 6, 14, 5
4. Estimate the mean for 0–10, 10–20, 20–30, 30–40 with frequencies 2, 7, 8, 3.

5. *How many three-digit codes from 1–4 contain at least one 1?*